1 Some anocdotes

The godfathers. In 1944 the book *Theory of Games and Economic Behavior* [4] von Neumann and Morgenstern introduce the expected utility hypothesis and prove that rational agents don't maximize expected dollars but maximize the expected utility. Markowitz put this in practice in 1952 by

2 Todos

- discrete time: random walk, markov process
- contineous time: brownian motion, stopping times, martingales
- Ito / stochastic calculus
- Risk neutral probability measure
- Risk neutral valuation, black scholes
- Tricks to avoid black scholes, law of unconciousness statisician (=fundamental theorem of asset pricing)
- Calll prices 'span' all derivative prices
- solving black scholes with neural networks
- the problem of trading frequency vs model accuracy

3 Risk neutral evaluation of derivatives

. A derivative (portfolio) is a collection of assets. The question is how to price such a derivative.

4 Heding

5 Lifting the assumptions

In the seminal paper [1] lift the assumptions of traditional models and rephrase the hedging problem into a optimization problem amenable to neural networks. While the original approach appreciates the recurrent nature of the problem it does not fully incorporate associated technologies (Transformers) or adapts reinforcement learning, which has been applied subsequently [3, 2]. Also note that diffusion or normalizing flow approaches are explored in this context.

References

[1] Hans Bühler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging, 2018.

[2] Ben Hambly, Renyuan Xu, and Huining Yang. Recent advances in reinforcement learning in finance. *Mathematical Finance*, 33(3):437–503, 2023.

[3] Shota Imaki, Kentaro Imajo, Katsuya Ito, Kentaro Minami, and Kei Nakagawa. No-transaction band network: A neural network architecture for efficient deep hedging. *The Journal of Financial Data Science*, 5(2):84–99, April 2023.

[4] John von Neumann and Oskar Morgenstern. *Theory of Games and Economic Behavior.* first edition, 1944.